What Privacy Actually Costs

An Accuracy-vs-Epsilon Curve for LoRA Fine-Tuning with DP-SGD

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The question

Differential privacy (DP) gives a provable guarantee: the trained model barely changes if any one training example is removed.

The guarantee is a single number, **ε**. Smaller ε = stronger privacy. But teams can't choose ε without knowing what each setting costs in accuracy and compute.

Setup: DP-SGD + LoRA in Opacus

SST-2 sentiment
67k training sentences

↓

RoBERTa-base, frozen
+ LoRA r = 8 on query/value: only ~0.7% of weights train

↓

Per-sample gradients
Opacus GradSampleModule, LoRA params only

↓

Clip each to norm C
bounds any one example's influence

↓

Add Gaussian noise σ·C
Poisson-sampled batch of 1,024

↓

Privacy accountant (RDP)
converts σ, steps, sampling rate → ε at $\delta = 10^{-5}$

Five noise levels, calibrated to target ε (3 epochs, 198 steps):

σ (noise)	0.576	0.736	0.954	1.287	1.918
ε	8	4	2	1	0.5

Why LoRA and DP fit together

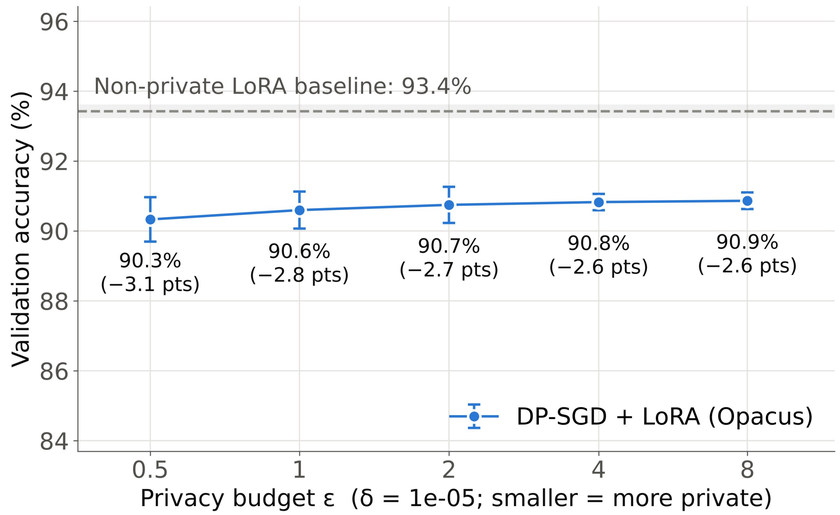
Opacus stores a gradient per example for every trainable weight. With LoRA that is ~0.9M weights instead of 125M, so DP-SGD fits on one GPU, and the noise is spread over far fewer dimensions.

Result: the price of each ε

93.4%
non-private LoRA baseline

90.3%
DP-LoRA at ε ≈ 0.5

−3.1 pts
the price of strong privacy



RoBERTa-base + LoRA on SST-2 (validation, n = 872). Mean ± std over 3 seeds (non-private and DP). Non-private LoRA baseline: 93.4%.

Takeaways

Privacy is cheap here, and nearly flat. With a tuned learning rate, DP-LoRA held 90.3–90.9% from ε ≈ 8 down to ε ≈ 0.5: 2.6–3.1 points below non-private.

The learning rate mattered more than ε. At LR 5e-3 the same runs lost 1.3–3.7 extra points, and the gap grew as ε shrank. Pricing ε without tuning LR misprices privacy.

Short pilots mislead. After 1 epoch, LR 2e-3 looked stuck at chance under heavy noise; after 3 epochs it was the best setting.

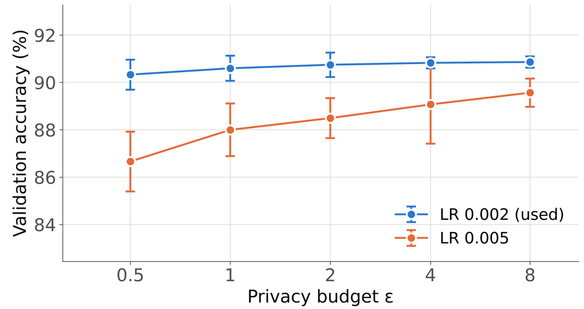
Practitioner checklist

- ✓ Freeze the base, train LoRA: per-sample gradients for only 0.7% of weights
- ✓ Tune LR at your target ε and full training length (DP wanted 4× the non-private LR)
- ✓ Fix δ < 1/N; report ε from the accountant you used (RDP here)
- ✓ Budget a sweep, not a single run, when you price privacy

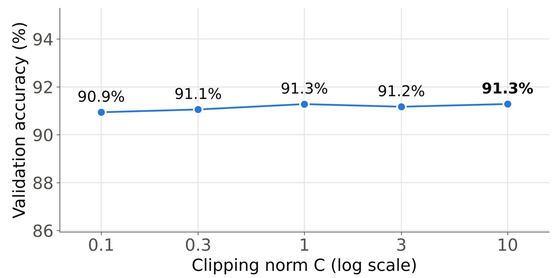
Friction points in practice

1 · BatchNorm is incompatible. It mixes samples within a batch, so per-sample gradients are undefined. Opacus rejects it; `ModuleValidator.fix()` swaps in GroupNorm. Transformers use LayerNorm and pass; LoRA on CNN/vision backbones hits this.

2 · DP needs its own learning rate, and it depends on ε. LR 5e-3, chosen by a 1-epoch pilot, loses 1.3–3.7 points (3-seed means), and the gap grows as ε shrinks.



3 · Clipping norm C barely mattered. At LR 2e-3 and ε ≈ 2, a 100× range of C (0.1–10) kept accuracy within 0.4 points.



4 · The real cost is the sweep. A DP run took about the same time as a non-private one (6.3 vs 6.1 min on 1× L40S, 7.7 vs 7.5 GB). The whole study, 55 runs with pilots and re-runs, was ≈ 5 GPU-hours (≈ $6).

Code, configs & raw results
github.com/vaibhavtupe/dp-lora-epsilon-curve

Compute: 1x NVIDIA L40S (Crusoe Cloud). Opacus · PyTorch · PEFT · HF Transformers.

Refs: Abadi et al. 2016 (DP-SGD); Mironov 2017 (RDP); Yu et al. 2022 (DP fine-tuning of LMs); Hu et al. 2022 (LoRA).